

Approaches for the discrimination of suppressive soils for *Pythium irregulare* disease



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ABSTRACT

The study of soil suppressiveness is of great interest in the field of sustainable agriculture. In the current study, eight soils located in the Region of Murcia (Spain) were sampled, evaluated and categorized according to their suppressiveness against *Pythium irregulare* in baby-leaf lettuce plants. We analyzed the chemical and biological properties as well as the bacterial and fungal community by sequencing 16S rRNA and ITS genes. We also attempted a non-targeted metabolomics analysis of the soil. The differences between the suppressive and non-suppressive soils were mainly associated with abiotic properties, such as pH, electrical conductivity (CE) and total organic carbon. Yet we also found links to biotic properties: several bacteria and fungi were found in suppressive soils (S) that were less abundant in the medium- (MS) or non-suppressive soils (NS). The differences observed in this study could play a decisive role against *P. irregulare*, and the data found could contribute significantly to the eventual control of this lettuce pathogen.

1. Introduction

Pythium root rot and damping off are common plant diseases that impact crop yields worldwide. Numerous *Pythium* spp., pathogenic fungus-like oomycetes, have been identified as causative agents of disease conditions in commercial crops. These pathogens are ubiquitous in soil environments and readily spread *via* airborne and fomite transmission (Sutton et al., 2006). Pathogenic strains have been isolated from greenhouse roof dust (Sanchez et al., 2001) and irrigation waters (Zappia et al., 2014), and this pathogen has caused plant mortality rates of up to 60% during epidemics (González et al., 2004; Serrano et al., 2008).

Soil suppressiveness is defined as the ability of a soil to prevent/suppress disease in the presence of a pathogen, a suitable host plant and favorable climatic conditions (Weller et al., 2002; Mazzola et al., 2004). This suppressiveness may be mediated by abiotic mechanisms, such as physicochemical properties, or it may directly or indirectly be a function of soil microorganisms. Soil microbes and phytopathogens share a common rhizosphere, and their interaction prior to crop sowing subsequently influences both plant growth and crop productivity (Penton et al., 2014).

Both general and specific suppression have been described. General suppression is related to the total microbial biomass and activity in a given soil, while specific suppression is related to the effects of individual or select groups of microorganisms during some stage in the life cycle of the pathogen (Weller et al., 2002). The management of the biologically complex components that contribute to disease-suppressive soils in agricultural production systems is often a prominent goal of research programs.

The soil microbial community is complex and requires new technologies like omics to characterize its composition and potential functionality. One common approach is to analyze the constituent microbial genomes through metagenomics (Campos et al., 2016; Löbmann et al., 2016). Metabolomics determines by-products released into the soil (Aguilar-Pulido et al., 2016) and has shown promise as one way to quickly quantitate the metabolites that microbes exude in response to climate stressors (Simpson et al., 2012)—such as changes in temperature, water and nutrient concentrations, or salinity (Rennenberg et al., 2009; Oikawa et al., 2011)—or due to the interactions of the pathogen with other microbes in the soil and the host plants. As a result, this method can be also a promising indicator of soil suppressiveness.

In this study, we look at different soil suppressiveness values against

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Table 1
Chemical and biological characteristics of the investigated soils. Data are represented as the mean ($n = 3$) $\pm$ standard deviation, different lower letters indicate significant differences between soils ($p \leq .05$). Upper letters indicate significant differences between suppressive groups (suppressive- middle- and non-suppressive, $p \leq .05$). Suppressive soils (S) S1 and S4; middle-suppressive soils (MS) S6 and S14; Non-suppressive soils (NS) S19, S22, S23 and S25.

Soils	Site	Coordinates	Plant types	pH	EC ($\mu\text{S cm}^{-1}$)	C _{org} (%)	N _t (%)	Dehydrogenase activity ($\mu\text{g INTF g}^{-1} \text{h}^{-1}$)
Forest								
S14	Revoladores	38.063611, -2.296667	Quercus rotundifolia	7.47 $\pm$ 0.02f B	357.66 $\pm$ 1.53a B	8.42 $\pm$ 0.03a A	0.47 $\pm$ 0.01a A	5.07 $\pm$ 0.44a A
S19	Burete	38.016669, -1.803889	Juniperus sp.	7.77 $\pm$ 0.07d B	137.00 $\pm$ 1.00e B	3.24 $\pm$ 0.02c AB	0.20 $\pm$ 0.02d A	3.62 $\pm$ 0.54b A
S23	Burete	38.016669, -1.803889	Quercus coccifera	7.57 $\pm$ 0.03e B	262.00 $\pm$ 2.64c B	3.97 $\pm$ 0.02b AB	0.34 $\pm$ 0.01b A	4.83 $\pm$ 0.15ab A
Shrublands								
S25	Abanilla	38.203611, -1.088333	Xerophytic shrub.	7.91 $\pm$ 0.02c B	131.23 $\pm$ 1.07ef B	0.83 $\pm$ 0.02e AB	0.16 $\pm$ 0.02de A	3.71 $\pm$ 0.85b A
S22	Burete	38.016666, 1.803888	Macrochloa tenacissima	7.80 $\pm$ 0.01d B	120.40 $\pm$ 1.51f B	3.24 $\pm$ 0.02c AB	0.28 $\pm$ 0.01c A	4.20 $\pm$ 0.50ab A
Agriculture soils								
S1	Cartagena	37.678058, -0.932593	Monoculture > 10 years	8.04 $\pm$ 0.02b A	368.66 $\pm$ 1.01a A	0.84 $\pm$ 0.01e B	0.13 $\pm$ 0.02ef B	0.75 $\pm$ 0.10c B
S4	Cartagena	37.7375278, -0.982011	Lettuce baby-leaf Monoculture > 10 years Lettuce baby-leaf	8.28 $\pm$ 0.02a A	325.00 $\pm$ 4.35b A	0.59 $\pm$ 0.05f B	0.10 $\pm$ 0.02f B	0.68 $\pm$ 0.13c B
S6	Cartagena	37.687017, -0.926771	Monoculture > 10 years Olea Europea	7.92 $\pm$ 0.02c B	248.00 $\pm$ 2.64d B	1.21 $\pm$ 0.01d A	0.17 $\pm$ 0.02d A	1.10 $\pm$ 0.04c A

Pythium irregulare and link these responses to the main parameters obtained from 16S rRNA and ITS community analyses, metabolome profile approaches and chemical and biological properties.

2. Materials and methods

2.1. Soil sampling

We collected eight soils (S1, S4, S6, S14, S19, S22, S23, and S25) from the Murcia Province in Spain with different plant types classified according to their soil type: Forest, Shrubland or Agricultural (Table 1). At each sampling site (S number), we chose one field for sampling. Each site was divided into three areas of approximately 10x20m, which were sampled separately as replicates for that site. The soil was sampled to a depth of 10–20 cm. Soil cores were taken in a W-pattern, and 10 soil cores were taken from each replicate area and mixed thoroughly. Soil samples were processed and stored at 4 °C and –20 °C for further analysis.

We measured the pH and EC in a 1/5 (w/v) aqueous soil extract using a pH meter (Crison mod. 2001, Barcelona, Spain) and determined the total organic carbon (C_{org}) and total nitrogen (N_t) using an Elemental Analyzer (C/N Flash EA 112 Series-Leco Truspec).

2.2. Pathogen strain isolation and preparation for the bioassay

Pythium irregulare (*P. irregulare*) was isolated from lettuce plants showing disease symptoms in a lettuce field. The *P. irregulare* inoculum was produced by transferring one agar plug (5 mm) of 2-day-old mycelia onto potato dextrose agar medium (PDA, Sharlau, Spain), which was autoclaved at 121 °C for 20 min and amended with 100 mg L⁻¹ sterile streptomycin. The pathogen solution for infection was obtained by recovering mycelia from a Petri dish in 50 mL of filter-sterilized distilled water using a glass spreader.

2.3. Soil suppressiveness assay

Pot experiments were performed by placing 20 g of soil and 5 red baby-leaf lettuce seeds in a 10 mL pot, with 30 pots for each soil (Antoria RZ variety, Rijk Zwaan, De Lier, Netherlands). Fifteen pots per soil were inoculated with *P. irregulare*. The inoculation dose was 10⁵ cfu g⁻¹ soil for the “pathogen treatment”. The other 15 pots with no pathogen inoculation were used as the “control treatments”. Germination was carried out in a germination chamber at 18 $\pm$ 1 °C and 90–95% relative humidity for two days. Once the seeds had germinated, the pots were placed randomly in a growth chamber with the following settings: 16 h, 24 °C and 70% humidity for the daylight conditions and 8 h, 18 °C and 70% humidity for the nighttime conditions. The seedlings were periodically watered, and 5 mL of nutrition solution (0.495 g L⁻¹ NO₃ NH₄; 0.33 g L⁻¹ KNO₃; 0.1275 g L⁻¹ PO₄H₂NH₄) was added once a week per pot. Soil suppressiveness was measured 15 days after sowing (das) and was calculated as the ratio between the average biomass of seedlings $\times$ the percentage of seed germination in the pathogen treatment and the average biomass of seedlings $\times$ the percentage of seed germination in the control treatments multiplied by 100.

2.4. Soil microbial community

2.4.1. DNA extraction, PCR and sequencing

Soil DNA extraction was performed using 0.5 g of the soil samples with a DNeasy PowerSoil Kit (100 preps, QIAGEN) following the manufacturer's instructions. The DNA was subjected to electrophoresis in 1.5% (w/v) agarose gels stained with ethidium bromide and visualized under UV light.

PCR amplification of the V4 region of bacterial 16S rDNA was performed using the barcoded primers 515F and 806R (Caporaso et al., 2012) in three PCR reactions per sample as described previously

(Žiřčáková et al., 2016). The PCR reactions contained 2.5 μL of $10\times$ buffer for DyNAzyme DNA Polymerase; 1.5 μL of BSA (20 mg mL^{-1}); 1 μL of each primer (0.01 mM); 0.5 μL of PCR Nucleotide Mix (10 mM each); 0.75 μL of polymerase (DyNAzyme II DNA polymerase 2 U/ μL + PFU 200 μL D + 6.6 μL PFU); 1 μL of template DNA; and 17.75 μL of water. The cycling conditions were as follows: 94°C for 4 min, 35 cycles of 94°C for 45 s, 50°C for 1 min, 72°C for 75 s, and a final extension at 72°C for 10 min. A partial region of the ITS1 region of the fungal ribosome encoding genes was amplified with a pair of primers, ITS7-ITS4 (Ihrmark et al., 2012), using the same PCR conditions for bacterial amplification. The cycling conditions were as follows: 94°C for 5 min, 35 cycles of 94°C for 30 s, 56°C for 30 s, 72°C for 30 s, and a final extension at 72°C for 7 min. The sequencing of bacterial and fungal amplicons was performed on Illumina MiSeq, and the sequences were generated with a MiSeq Reagent Kit v2 in paired-end mode with sizes of 251 base pairs in the C4SYS facility at the Institute of Microbiology of the CAS, Prague, Czech Republic.

2.4.2. Sequence data processing and analysis

The amplicon sequencing data were processed using the pipeline SEED 2 (Vetrovsky et al., 2018). Briefly, pair-end reads were merged using fastq-join (Aronesty, 2013), and whole amplicons were processed. Chimeric sequences were detected using Usearch 7.0.1090 (Edgar, 2010) and deleted. UPARSE implemented within Usearch (Edgar, 2013) was used for clustering at a 97% similarity level. Consensus sequences were constructed for each cluster, and the closest hits at a genus or species level were identified using BLASTn against the RDP (Cole et al., 2014) and Genbank databases. The full taxonomy of each best hit was retrieved from GenBank. Sequences identified as non-bacterial or non-fungal and singletons were excluded from further analyses. The pipeline SEED 2 (Vetrovsky et al., 2018) was used for data pre-processing, and the sequences were uploaded to MGRAST-ID (mgm4826506.3/mgm4860533.3).

The Shannon diversity index was calculated with the “Vegan” package version 2.5–5 using an OTU table (sample) (Oksanen et al., 2017).

2.5. Metabolite analysis

2.5.1. Metabolite extraction

One gram of the soil samples was added to 4 mL water (HPLC grade), shaken (2000 rpm) at 4°C for 1 h, and then centrifuged (3220 RCF) at 4°C for 15 min. The supernatant was discarded, and 4 mL ethyl acetate was added. The re-suspended material was agitated at 4°C for 1 h, sonicated at 20°C for 10 min and centrifuged (3220 RCF) at 4°C for 15 min. The supernatant was collected and dried in a SpeedVac centrifugal vacuum concentrator (Thermo) for 120 min with no heat applied. The dried material was re-suspended in 0.15 mL of LC-MS grade methanol containing additional internal standards [1 $\mu\text{g mL}^{-1}$ 2-amino-3-bromo-5-methylbenzoic acid ($5\text{ }\mu\text{g mL}^{-1}$), 3,6-dihydroxy-4-methylpyridazine, ($5\text{ }\mu\text{g mL}^{-1}$), L-phenylalanine- $^{13}\text{C}_9$, ^{15}N , ($10\text{ }\mu\text{g mL}^{-1}$) L-lysine-4,4,5,5-d $_4$, ($10\text{ }\mu\text{g mL}^{-1}$), benzoic acid-2,3,4,5,6-d $_5$, ($0.2\text{ }\mu\text{g mL}^{-1}$) 9-anthracene carboxylic acid]. The resuspensions were bath-sonicated at 20°C for 10 min, vortexed briefly two times and centrifuged at 4°C for 5 min (3220 RCF). The supernatant was filtered (0.22 μm PVDF) before being transferred to glass vials for LCMS analysis.

2.5.2. Q-TOF-MS Analysis and spectra processing

Metabolite extracts were chromatographically separated using a Zorbax Eclipse Plus C18 column (Agilent, 959757-902, $50\times 2.1\text{ mm}$, $1.8\text{ }\mu\text{m}$) on an Agilent 1290 LC system. Metabolites were analyzed on a downstream Q Exactive Hybrid Quadrupole-Orbitrap Mass Spectrometer (Thermo Scientific) equipped with a HESI-II source probe using Full MS with Data-Dependent tandem MS. The LCMS parameters are available in the supplementary material in Tables S1–S3. Each sample was injected into two sequential injections, first for positive

mode analysis and then for negative mode analysis. Sample injections were randomized with solvent blank injections between each one; internal and external standards were used for quality control purposes. For untargeted metabolite analysis, features were detected using MZMine version 2.24 (<https://doi.org/10.1093/bioinformatics/btk039>).

2.6. Data analysis

One-way ANOVA followed by Tukey's post hoc test ($p < .05$) was used to compare the data obtained from the physico-chemical, chemical and biological soil properties. We also calculated Pearson correlations among the different parameters, and, in order to compare the phylogenetic and metabolomic data of the microbial communities between soils, principal component analysis (PCA) was carried out using SPSS 23.0 software (SPSS Inc., Chicago, IL, USA).

3. Results

3.1. Soil suppressiveness

The suppressiveness of the soils studied against *P. irregulare* is shown in Fig. 1. There are significant differences between soils except for soils S19 and S23, which showed similar suppressiveness. Soils were categorized as suppressive (S) (S1 and S4; 92%–93%); middle-suppressive (MS) (S6 and S14; 82%–77%); or non-suppressive (NS) (S19, S22, S23 and S25; 33%–44%).

3.2. Soil physico-chemical, chemical and biological properties

The pH, total organic carbon (C_{org}), total nitrogen (N_t) and dehydrogenase activity showed significantly different values according to the soil suppressiveness against *P. irregulare* (Table 1). The pH and EC had significantly higher values ($F = 14.328$, $F = 10.665$; $P \leq .05$), respectively, in S soil than in MS and NS soils. The C_{org} and N_t values, however, were significantly higher in NS and MS soils than in S soil ($F = 5.577$, $F = 6.838$; $P \leq .05$), respectively. Dehydrogenase activity was also significantly higher in NS and MS soils ($F = 16.289$; $P \leq .05$).

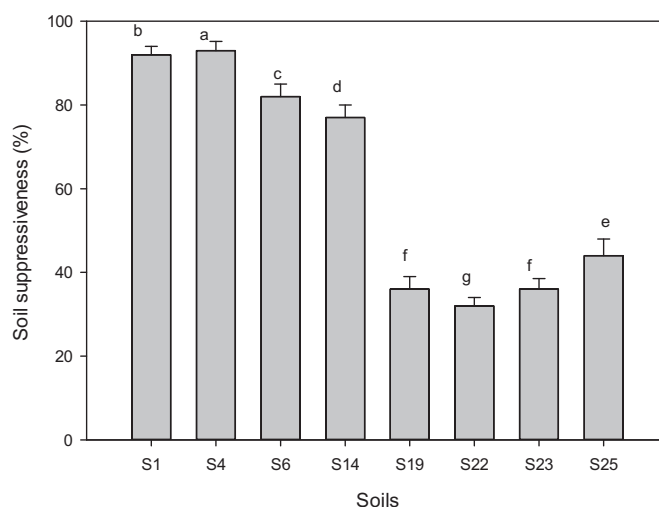


Fig. 1. Soil suppressiveness against *Pythium irregulare* (%). (Bars indicate mean values of different suppressiveness soils against *Pythium irregulare*. Error bars represents standard deviation and different letters indicate significant differences between soils ($p < .05$)); Suppressiveness soils (S) S1 and S4 Agriculture soils (Monoculture, lettuce baby-leaf); Middle-suppressive soils (MS) S6, Agriculture soil (Monoculture, *Olea Europea*) and S14, Forest (*Quercus rotundifolia*); Non-suppressive soils (NS) S19, Forest (*Juniperus sp.*); S22, Shrublands (*Macrochloa tenacissima*); S23, Forest (*Quercus coccifera*); S25, Shrublands (*Xerophytic shrub*).

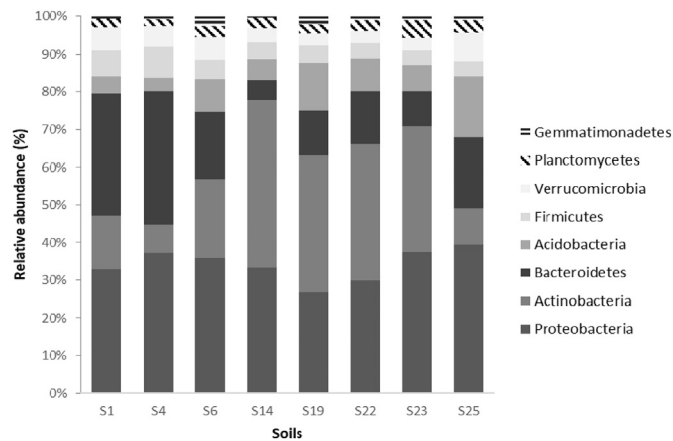


Fig. 2. Relative abundance of most abundant bacterial phyla in the investigated soils (%). Suppressive soils (S) S1 and S4 Agriculture soils (Monoculture, lettuce baby-leaf); Middle- suppressive soils (MS) S6, Agriculture soil (Monoculture, *Olea Europea*) and S14, Forest (*Quercus rotundifolia*); Non-suppressive soils (NS) S19, Forest (*Juniperus* sp.); S22, Shrublands (*Macrochloa tenacissima*); S23, Forest (*Quercus coccifera*); S25, Shrublands (*Xerophytic shrub*).

than in S soil.

3.3. Bacterial community diversity and composition

The Shannon diversity index of the bacterial community was significantly different between soils (Table S4), and NS soil showed higher diversity than the others (S and MS). The total number of OTUs was 8668 (Table S5). Among the most abundant phylum (relative abundance > 1%) (Fig. 2), Proteobacteria was the most abundant (34% average of all samples), followed by Actinobacteria (25%), Bacteroidetes (18%) and Acidobacteria (8%) (Fig. 2). Principal component analysis (PCA) explaining 54.69% of the soil bacterial phyla community showed S soils clustered separately from the other soils along PC1 (Fig. 3a). The principal phyla involved in this separation were Bacteroidetes, which were significantly ($P \leq .05$) higher in S soils than in MS or NS soils, and Actinobacteria, which were significantly ($P \leq .05$) lower in S soils than in MS or NS soils.

Forty-eight orders with > 0.5% relative abundance were observed. PCA revealed that, at the order level, S soils clustered separately from MS and NS soils at the PC1 (34.9%) axis (Fig. 3b). Sixteen orders were principally involved in this separation. Among these orders, Cytophagales, Clostridiales, Optitutales, Cellvibrionales, Pseudomonadales, Xanthomonadales and Verrucomicrobiales showed significantly ($P \leq .05$) higher abundance in S soils than in the other soils, while Corynebacteriales, Gaiellales, Solirubrobacteriales and Rhizobiales showed significantly ($P \leq .05$) lower abundance in S soils than in MS and NS soils (Fig. 3b).

The most abundant genera (> 2% relative abundance) are shown in Table S6. The differences between suppressive soils were particularly notable in the genera *Flavisolibacter*, *Flexibacter*, *Pontibacter*, *Bacillus* and *Pseudomonas*, which were significantly ($P \leq .05$) higher in S soils than in MS and NS soils.

3.4. Fungal community composition

The Shannon diversity index of the fungal community was also significantly different between soils (Table S4), and out of the different NS soils, S19 and S23 showed the highest diversity index. The total number of OTUs was 4893 (Table S5). The most abundant phyla were Ascomycota (51% average of all samples), followed by Basidiomycota (21%) (Fig. 4). PCA explaining 65.75% of the fungal phyla community showed a separation on PC1 (40.73%) between S and MS soils on one hand, and NS soils on the other. Furthermore, the relative abundance of

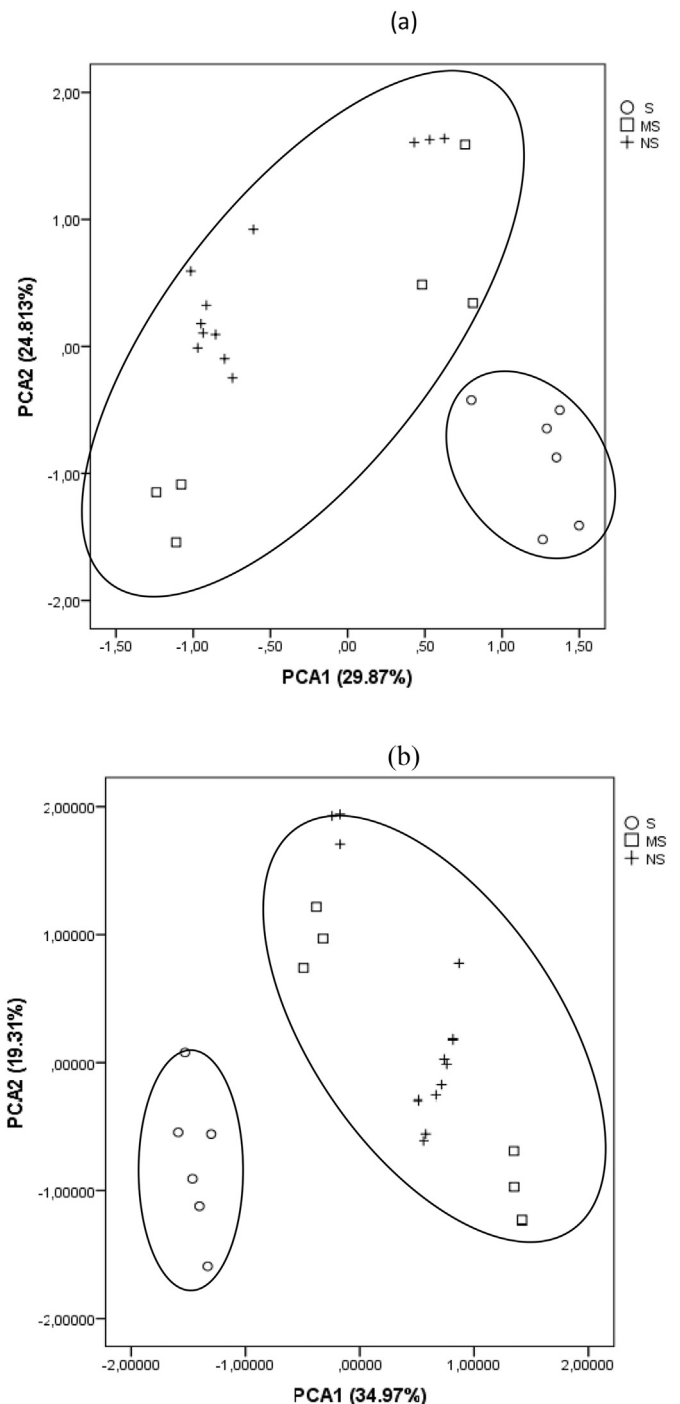


Fig. 3. Principal Component analysis (PCA) of bacterial a) phyla and b) orders in the investigated soils. Circles group the different soils according to the PCA. Suppressive soils (S) S1 and S4, Agriculture soils (Monoculture, lettuce baby-leaf); Middle- suppressive soils (MS) S6, Agriculture soil (Monoculture, *Olea Europea*) and S14, Forest (*Quercus rotundifolia*); Non-suppressive soils (NS) S19, Forest (*Juniperus* sp.); S22, Shrublands (*Macrochloa tenacissima*); S23, Forest (*Quercus coccifera*); S25, Shrublands (*Xerophytic shrub*).

Ascomycota was significantly higher ($F = 7.007$; $P \leq .05$) in S and MS soils than in NS soils (Fig. 5a). Basidiomycota, on the other hand, was significantly lower ($F = 20.975$; $P \leq .05$) in S and MS than in NS soils.

In the fungal community, 18 orders with > 0.5% relative abundance were identified. PCA explained 52.94% of the total fungal community and showed differences between S and MS soils compared to NS soils on PC1 (37.35%) (Fig. 5b). The principal orders involved were

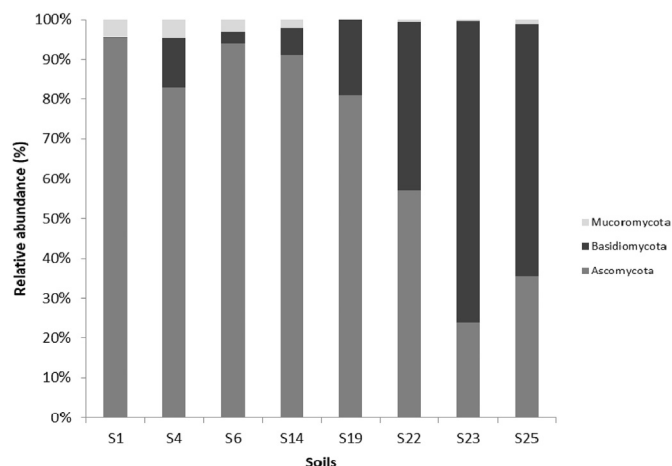


Fig. 4. Relative abundance of most abundant fungal phyla in the investigated soils (%). Suppressive soils (S) S1 and S4, Agriculture soils (Monoculture, lettuce baby-leaf); Middle- suppressive soils (MS) S6, Agriculture soil (Monoculture, *Olea Europea*) and S14, Forest (*Quercus rotundifolia*); Non-suppressive soils (NS) S19, Forest (*Juniperus sp.*); S22, Shrublands (*Macrochloa tenacissima*); S23, Forest (*Quercus coccifera*); S25, Shrublands (*Xerophytic shrub*).

Glomerales and Hypocreales, which showed a higher abundance in S and MS soils ($F = 4.424$; $F = 16.40$; $P \leq .05$, respectively). Sordariales and Mortierellales ($F = 32.41$; $F = 17.708$; $P \leq .05$, respectively) showed the highest abundance in S soils followed by MS and NS soils.

The most abundant genera ($> 2\%$ relative abundance) are shown in Table S7. The genera *Aspergillus*, *Fusarium*, *Mortierella*, *Acremonium*, *Alternaria*, *Chaetomium* and *Humicola* were significantly ($P \leq .05$) more abundant in S soils than in MS and NS soils.

3.5. Soil metabolites

Soil metabolite extraction was carried out using ethyl acetate to improve the secondary metabolites with antimicrobial activity (Skariyachan et al., 2013; Ripa et al., 2010). The results of LC-MS were analyzed on positive mode, and 55 features were identified in all soil samples. PCA revealed two components that accounted for 40.4% of the variance. S was separated along the C1 axis from MS and NS (Fig. 6), due to features with m/z 313142, 345168, 521286, 519270, 4693302, 233087, 497288 that were significantly ($P \leq .05$) greater in S soils than in the others.

3.6. Correlations

Pearson correlation analysis showed a positive correlation between soil suppressiveness ($p \leq .01$), pH and EC and a negative correlation between soil suppressiveness and dehydrogenase activity.

Furthermore, the pH showed a positive ($p \leq .01$) correlation with Bacteroidetes and a negative correlation ($p \leq .01$) with Actinobacteria, while Corg showed a positive ($p \leq .01$) correlation with Actinobacteria and a negative correlation ($p \leq .01$) with Bacteroidetes. Dehydrogenase showed a positive ($p \leq .01$) correlation with Actinobacteria and a negative correlation ($p \leq .01$) with Bacteroidetes.

4. Discussion

Disease-suppressive soils are already an important resource that can be exploited for controlling soil-borne pathogens (Mendes et al., 2011; Latz et al., 2016). Such soils offer plants effective protection against infection by soil-borne pathogens while minimizing pesticide and fertilizer input (Exposito et al., 2017). Soil suppressiveness is interactively shaped by various factors, such as abiotic properties, management practices, and the structure and function of indigenous microorganisms

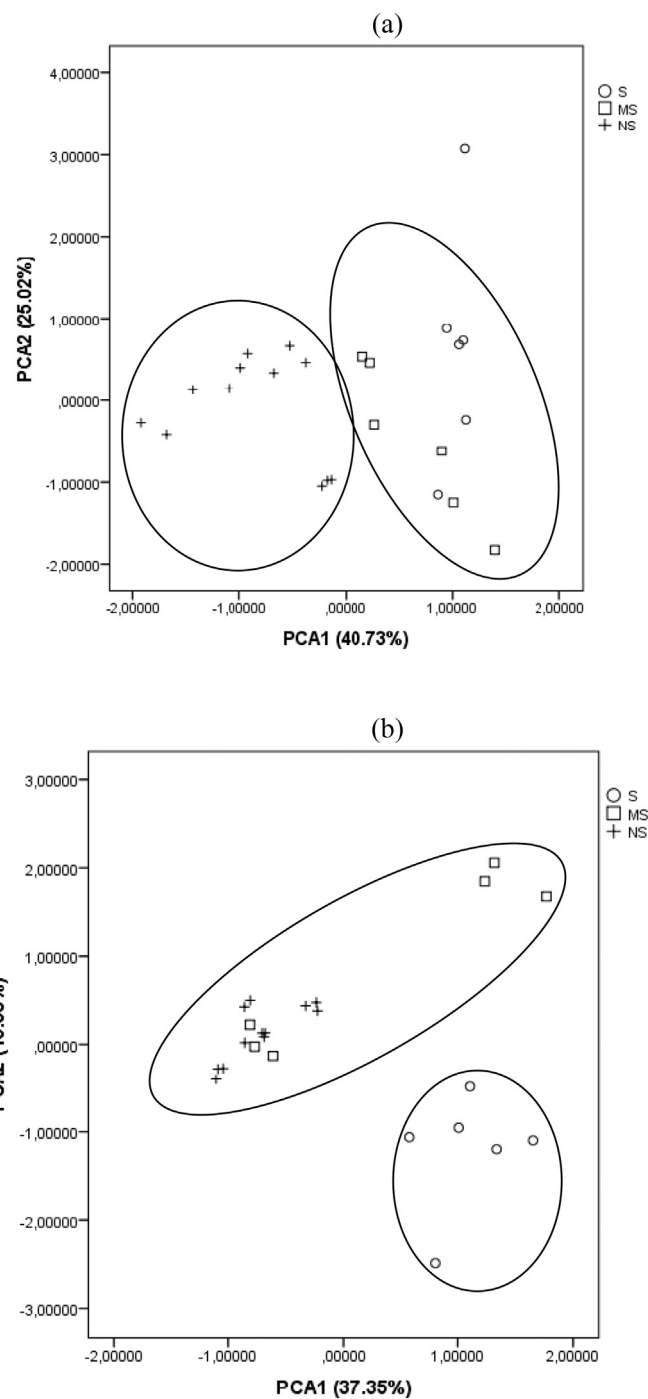


Fig. 5. Principal component Analysis (PCA) of fungal a) phyla and b) orders in the investigated soils. Circles group the different soils according to the PCA Suppressive soils (S) S1 and S4 Agriculture soils (Monoculture, lettuce baby-leaf); Middle- suppressive soils (MS) S6, Agriculture soil (Monoculture, *Olea Europea*) and S14, Forest (*Quercus rotundifolia*); Non-suppressive soils (NS) S19, Forest (*Juniperus sp.*); S22, Shrublands (*Macrochloa tenacissima*); S23, Forest (*Quercus coccifera*); S25, Shrublands (*Xerophytic shrub*).

(Katan, 2017). In this study, we found that *P. irregulare*-soil suppressiveness (S1 and S4) is associated with a set of abiotic and biotic properties that are linked, either directly or indirectly, to soil community composition. The chemical properties in the suppressive soils, such as the lower levels of organic carbon and total nitrogen, agree with previous findings in which soil suppressiveness occurs less frequently under high-nutrient conditions (Knudsen et al., 2002; Tamm et al.,

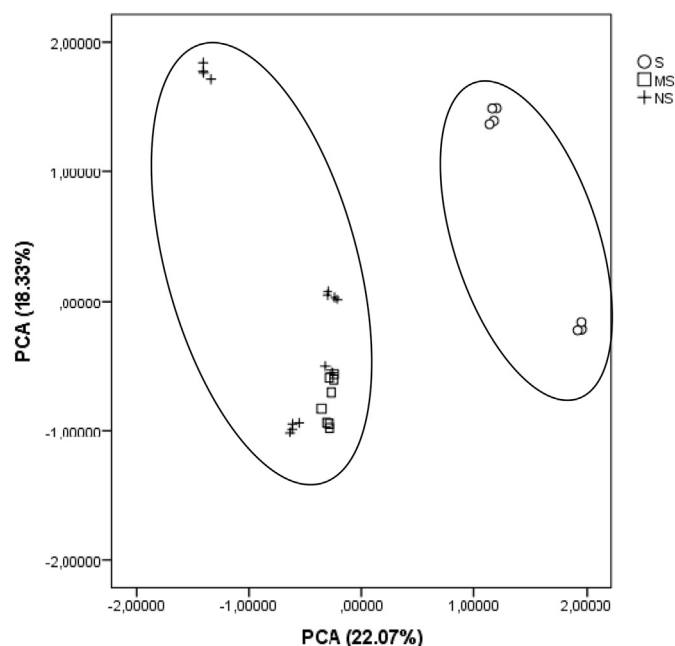


Fig. 6. Principal component analysis (PCA) of metabolites analyzed from the investigated soils. Circles group the different soils according to the PCA. Suppressive soils (S) S1 and S4 Agriculture soils (Monoculture, lettuce baby-leaf); Middle- suppressive soils (MS) S6, Agriculture soil (Monoculture, *Olea Europea*) and S14, Forest (*Quercus rotundifolia*); Non-suppressive soils (NS) S19, Forest (*Juniperus* sp.); S22, Shrublands (*Macrochloa tenacissima*); S23, Forest (*Quercus coccifera*); S25, Shrublands (*Xerophytia* shrub).

2010). High nutrient levels promote plant pathogen growth (Miller and Rosen, 2005) and make plants more susceptible to pathogens. Biotic properties, on the other hand, indicated the low microbial activity of the suppressive soils in our study, which disagrees with other studies in which high microbial activity has been correlated with the suppression of soil-borne disease (Bonanomi et al., 2010; Blaya et al., 2015). It is possible that soils respond to particular types of disease suppression (Schlatter et al., 2017). In other words, suppressiveness is initiated and sustained by crop monoculture, by growing crops susceptible to a given disease, and/or by adding inoculum of a pathogen to a soil in which disease outbreak is required for the onset of suppressiveness (Weller et al., 2002; Haas and Defago, 2005).

High pH has been linked to a higher abundance of Bacteroidetes (Fierer et al., 2012; Wang et al., 2017; Blaya et al., 2016), as we observed in our suppressive soils (S). These soils had a higher abundance of microorganisms related to Bacteroidetes (*Flavisolibacter*, *Flexibacter* and *Pontibacter*), Firmicutes (*Bacillus*) and gamma-proteobacteria (*Pseudomonas*). The latter two are particularly important for soil suppressiveness (Weller et al., 2002; Mendes et al., 2011). The relationship between the genera *Flavisolibacter*, *Flexibacter* and *Pontibacter* and soil suppressiveness has been less studied. Minna et al. (2008) observed that *Flexibacter* in combination with *Pseudomonas*, a key specie in the suppression of fungal growth, contributed to soil fungistasis. Bonanomi et al. (2018) showed that the genera *Flavisolibacter* and *Pseudomonas*, among others, positively correlated with disease suppression. *Pseudomonas* species are associated with antibiotic production, like the 2,4-diacetylphloroglucinol (DAPG) and phenazines (PHZ) produced by several strains of *Pseudomonas* (fluorescent) (Raaijmakers and Mazzola, 2012), an underlying specific mechanism of suppressiveness among other mechanisms (Alabouvette et al., 2009) against *P. irregulare* (de Souza et al., 2003). However, Imperiali et al. (2017) observed that high numbers of these bacteria often support only low levels of antimicrobial gene expression and *vice versa*, and that their impact can be influenced by multiple soil factors. *Bacillus* not only control a wide range of plant

pathogens, but they also rapidly and aggressively colonize the root system and enhance plant growth and yield by direct and indirect plant growth-promoting (PGP) activities (Kumar et al., 2012).

In our suppressive soils (S), we found a higher abundance of Ascomycota (*Acremonium*, *Humicola*, *Chaetomium*) and Mucoromycota (*Mortierella*), in accordance with the literature (Gulden and Vessey, 2000; Gongshuai et al., 2018; van Bruggen and Semenov, 2000; Yuliar et al., 2015; Lee et al., 2017). *Acremonium* spp. are a well-known biological control agent used to protect against several plant diseases (Lee et al., 2017). *Chaetomium* and *Acremonium* have been defined as beneficial fungi that enhance soil quality and inhibit soil pathogens (Soytong et al., 2001; Anisha and Radhakrishnan, 2015; Wang et al., 2017). Furthermore, Rodríguez et al. (2015) showed higher abundance of *Humicola* in suppressive soils due to their antifungal activity. *Mortierella* sp. are also considered to be beneficial fungi against different pathogens in plants by inducing systemic resistance and promoting plant growth, competition for nutrients and/or antibiotic activity (Gulden and Vessey, 2000; Gongshuai et al., 2018).

It is important to highlight the presence of different fungal pathogens like *Alternaria* sp., *Aspergillus* sp. and *Fusarium* sp. in suppressive soils. Such pathogens can compete for space and limited resources in the plant environment, and this competition is a mechanism of disease suppression (Bonanomi et al., 2010). Also, some non-pathogenic strains like *Fusarium* sp. could show antagonistic activity against fungal pathogens.

Overall, in accordance with other studies, our results suggest that disease-suppressive soils cannot be ascribed to a single bacterial/fungal taxon or group and that suppressiveness is most likely governed by an enriched microbial consortia of fungal and bacterial networks (Mendes et al., 2011; Xiong et al., 2017). This makes sense considering that a highly connected network may provide resistance to disturbances like pathogen attacks (Scheffer et al., 2012). This knowledge could help in engineering the soil microbiome as a promising means for protecting plant health (Mueller and Sachs, 2015).

After PCA, metabolomic analysis of the soils studied indicated a separation between suppressive soils (S) on the one hand and middle- and non-suppressive soils (MS and NS) on the other hand. Similar results were observed by Hayden et al. (2019), who found a separation between suppressive and non-suppressive soils using PCA and an antimicrobial secondary metabolite (macrocarpal) as the most abundant compound in suppressive soils. Mehta et al. (2016) also found several volatile organic acid sterols considered toxic to microorganisms in disease-suppressive soils.

5. Conclusions

Our results indicate that the interaction among abiotic and biotic properties are related to the level of soil suppressiveness against *P. irregulare*. Both suppressive soils (S and MS) showed higher electrical conductivity than NS and lower levels of organic carbon, which could establish a certain presence of fungi and bacteria that may or may not be interrelated. However, it is important to remark that abundance of these fungi and bacteria, in turn, help to make soils more resistant to pathogen attacks.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apsoil.2019.103439>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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